

$$\therefore E_1 A = \begin{bmatrix} a & b & c \\ d & e & f \\ g-4a & h-4b & i-4c \end{bmatrix}$$

$$E_2 A = [E_{21} \ E_{22} \ E_{23}] \quad (\text{By defn. of matrix multiplication})$$

$$E_{21} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ d \\ g \end{bmatrix} = a \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + g \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad (\text{By defn. of } Ax)$$

$$= \begin{bmatrix} 0 \\ a \\ 0 \end{bmatrix} + \begin{bmatrix} d \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} \quad (\text{Scalar multiplication}) = \begin{bmatrix} d \\ a \\ g \end{bmatrix} \quad (\text{Vector addition})$$

$$E_{22} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} b \\ e \\ h \end{bmatrix} = \begin{bmatrix} e \\ b \\ h \end{bmatrix} \quad (\text{Similar calculation as } E_{21})$$

$$E_{23} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c \\ f \\ i \end{bmatrix} = \begin{bmatrix} f \\ c \\ i \end{bmatrix} \quad (\text{Similar calculation as } E_{21})$$

$$\therefore E_2 A = \begin{bmatrix} d & e & f \\ a & b & c \\ g & h & i \end{bmatrix}$$

$$E_3 A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = [E_{31} \ E_{32} \ E_{33}] \quad (\text{By defn. of matrix multiplication})$$

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} a \\ d \\ g \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + g \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} \quad (\text{By defn. of } Ax)$$

$$= \begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ d \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 5g \end{bmatrix} \quad (\text{Scalar multiplication}) = \begin{bmatrix} a \\ d \\ 5g \end{bmatrix} \quad (\text{Vector addition})$$

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} b \\ e \\ h \end{bmatrix} = \begin{bmatrix} b \\ e \\ 5h \end{bmatrix} \quad (\text{Similar calculation as } E_{31})$$

$$E_{33} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} c \\ f \\ i \end{bmatrix} = \begin{bmatrix} c \\ f \\ 5i \end{bmatrix} \quad (\text{Similar calculation as } E_{31})$$